

Early Spring: Integrating Forces

Based on the provided Python (e.g. `spring.py`), complete the following exercises. You will need to provide written answers to the questions (PDF) and separately submit code as two `.py` files with the names indicated below.

Note that cosmetic (visual) code improvements do not matter for this assignment (i.e. you can use `spring.py` or `spring_rot.py` as a starting point for your code as you prefer – see Code/Forces on avenue). The default rest length $L=15$ and the default spring constant $k=1$.

Exercise 1

Investigate the stability of the Euler integrator. If you increase the spring constant k too much the integrator will be unstable. At what value of k_{stable} does it explode for $k > k_{\text{stable}}$? Note that it may look visually strange for large k values just because of how fast it is going. That is not unstable. I am talking about actually crashing Python. You should hardcode the time step $dt=1/60$ to avoid lag or related issues affecting the test. Try and pin it down to ~ 2 figures. **Report the value of k_{stable} in your submission.**

Exercise 2

Add damping to the spring following the equations presented in class (Forces lecture) and test it out.

Exercise 3

Create a critically damped version of the spring code and demonstrate it in code. Critical damping is the smallest value of the damping coefficient so that the springs relax to the rest length without over-shooting even once.

As a set-up, have the two masses positioned at $x=-20$ and $x=20$ with no initial velocity. Hand this in as **`spring_critical.py`**

What value of the damping coefficient (D) was needed to get the critical damping result?

Note that the critical damping value may not be the same as for a single mass on a spring attached to a wall.

Exercise 4

Create an initial condition with the masses spinning about the initial centre of mass (e.g. about $x=0, y=50, z=0$ or similar). The centre of mass should not move. Keep the rest length at 15 and $k=10$. Determine the velocity of each particle (e.g. the initial velocity) such that the particles maintain a separation of 20 after the oscillations stop. Hand this code in as **`spring_20.py`**

What is the initial velocity to give a final steady separation of 20?

What is the physical explanation for the mass separation being larger than the rest length? (Don't just explain in words, include math in your answer)